

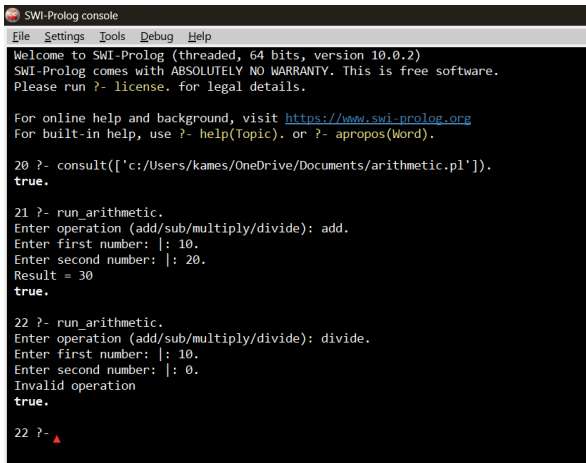
3-d Arithmetic in Prolog

PROGRAM:

```
add(X, Y, R) :- R is X + Y.
sub(X, Y, R) :- R is X - Y.
multiply(X, Y, R) :- R is X * Y.
divide(X, Y, R) :- Y \= 0, R is X / Y.

run_arithmetic :-
    write('Enter operation (add/sub/multiply/divide): '), read(Op),
    write('Enter first number: '), read(X),
    write('Enter second number: '), read(Y),
    Goal =.. [Op, X, Y, R],
    (call(Goal) -> format('Result = ~w~n', [R]) ; writeln('Invalid operation')).
```

OUTPUT:



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

20 ?- consult(['c:/Users/kames/OneDrive/Documents/arithmetic.pl']).
true.

21 ?- run_arithmetic.
Enter operation (add/sub/multiply/divide): add.
Enter first number: |: 10.
Enter second number: |: 20.
Result = 30
true.

22 ?- run_arithmetic.
Enter operation (add/sub/multiply/divide): divide.
Enter first number: |: 10.
Enter second number: |: 0.
Invalid operation
true.

22 ?- ▲
```